Machine Learning Fundamentals: A Practitioner's Guide

1. Introduction to Machine Learning

Machine learning (ML) is a subfield of artificial intelligence in which algorithms learn patterns from data rather than following explicitly programmed rules. The field encompasses supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning — each suited to different problem structures and data availability scenarios.

The modern ML workflow consists of five stages: data collection and cleaning, feature engineering, model selection and training, evaluation, and deployment. Neglecting any stage degrades overall system performance. In practice, data quality and feature design account for a greater share of model performance variance than architecture choice alone.

2. Supervised Learning Algorithms

Supervised learning trains a model on labelled examples to predict outputs for unseen inputs. Linear regression models continuous targets by fitting a hyperplane to minimise mean squared error (MSE). Logistic regression extends this to binary classification via the sigmoid activation, producing calibrated probability estimates.

Decision trees partition the feature space through recursive binary splits, choosing splits that maximise information gain or Gini impurity reduction. Ensemble methods — Random Forests and Gradient Boosted Trees (e.g., XGBoost, LightGBM) — combine many weak trees to achieve state-of-the-art performance on tabular data. Gradient boosting builds trees sequentially, each correcting the residual errors of its predecessors.

Support Vector Machines (SVMs) find the maximum-margin hyperplane separating classes in a transformed feature space defined by a kernel function. The radial basis function (RBF) kernel maps inputs to an infinite-dimensional space, enabling non-linear decision boundaries with computational efficiency via the kernel trick.

3. Neural Networks and Deep Learning

Artificial neural networks are composed of layers of interconnected nodes (neurons) that apply learned affine transformations followed by non-linear activation functions. Stacking many such layers — deep learning — enables the extraction of hierarchical feature representations directly from raw input data such as pixels, audio waveforms, or tokens.

Backpropagation computes the gradient of the loss function with respect to all network parameters by applying the chain rule through the computational graph. Stochastic gradient descent (SGD) and its adaptive variants — Adam, AdamW, RMSProp — use these gradients to iteratively update weights. Batch normalisation, dropout, and residual connections address training instability and vanishing gradient problems in very deep networks.

Transformer architectures, introduced in the 2017 paper 'Attention Is All You Need', have become the dominant paradigm for natural language processing. The self-attention mechanism computes pairwise relationships between all tokens in a sequence, enabling the model to capture long-range dependencies without the sequential bottleneck of recurrent networks.

4. Model Evaluation and Validation

Proper evaluation prevents over-fitting to the training distribution. The standard approach partitions data into training, validation, and test sets. The training set fits model parameters; the validation set guides hyperparameter selection; the test set provides an unbiased performance estimate. Data leakage — allowing information from the test set to influence any earlier stage — is a pervasive source of inflated benchmark results.

Classification metrics include accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC). For imbalanced datasets, accuracy is misleading; precision-recall curves and the Matthews Correlation Coefficient (MCC) provide more informative summaries. Regression metrics include MSE, RMSE, mean absolute error (MAE), and R^2.

k-Fold cross-validation partitions data into k equal folds, training k models each using a different fold as the validation set. The mean and standard deviation of fold scores estimate generalisation performance and its variance. Stratified k-fold preserves class proportions across folds, which is essential for imbalanced classification problems.

5. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines a dense retrieval system with a large language model (LLM) to ground generated responses in relevant source documents. A query encoder maps the user's question to a dense vector; an approximate nearest-neighbour index such as FAISS retrieves the top-k most similar document chunks; and the LLM generates an answer conditioned on both the query and the retrieved context.

Chunking strategy critically affects retrieval quality. Fixed-size character windows are simple but may split sentences mid-thought. Sentence or paragraph-level chunking respects linguistic boundaries but produces variable-length chunks. Recursive character text splitters attempt to split on paragraph, then sentence, then word boundaries, balancing coherence and uniformity. Chunk overlap — repeating tokens at boundaries — helps preserve context across adjacent chunks.

Embedding model choice affects both retrieval precision and computational cost. OpenAI's text-embedding-3-small model produces 1,536-dimensional vectors and offers a strong quality-cost balance for most RAG applications. Re-ranking with a cross-encoder after initial retrieval further improves precision by scoring query-chunk pairs jointly rather than independently.